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IDEA

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Accounting & Valuation | North America

AI Ecosystem: Charting Recent Trends

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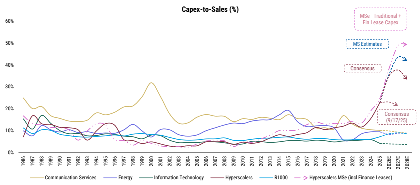
As hyperscaler capital intensity reaches more than 40% of sales, incremental capex revisions are now accompanied by capital raises and/or innovative financing. Long-term purchase commitments of ~$1tn and lease commitments >$800bn further support the AI buildout, but add off-BS operating leverage.

Key Takeaways

- Our analysts forecast capex-to-sales ratios of 36%, 44%, and 42% in 2026–28, well above the 32% dot-com fiber peak. Finance leases push headline numbers higher.
- AI is dominating overall investment, with AI-related spend expected to make up more than 50% of R1000 capex in 2026, of which hyperscalers contribute ~90%.
- Among major AI compute players, RPO has nearly tripled in the last year, reaching >$2tn as customers commit to longer-duration contracts.
- Depreciation is the next margin watch item, with MSFT, ORCL, META, and GOOGL depreciation is expected to exceed $520bn over three years.
- See full slide deck inside with our most popular AI charts updated for the latest SEC filings.

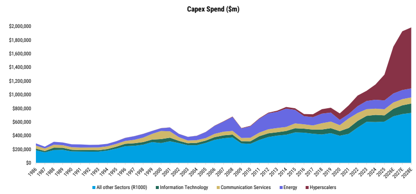
Please refer to the slides below for refreshed versions of our key charts from the past year, reflecting an updated view of the AI ecosystem.

Exhibit 1: MSe expects Hyperscaler capex/sales to surpass dot-com peaks reaching 44% in '27



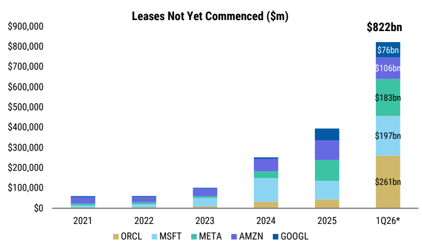
Note: The hyperscalers group includes AMZN, GOOGL, META, MSFT, & ORCL and are excluded from the other sector groups. Only MSFT and ORCL finance lease estimates are included. R1000 excludes Real Estate and Financials.
Source: Morgan Stanley Estimates and Factset.

Exhibit 2: Cons expects Hyperscaler capex to reach 45% of the R1000 in '27 and '28



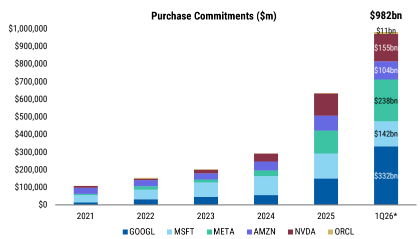
Note: The hyperscalers group includes AMZN, GOOGL, META, MSFT, & ORCL and are excluded from the other sector groups. R1000 excludes Real Estate and Financials. Estimates are based on consensus.
Source: FactSet.

Exhibit 3: Leases not yet started have hit
>$800bn across Hyperscalers...



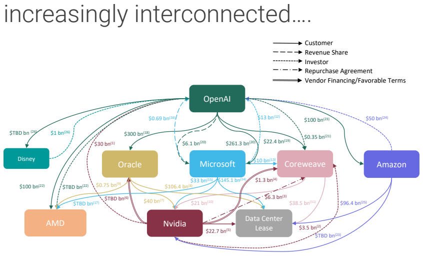
Note: Based on fiscal years. ORCL and MSFT 1Q26 = 3FQ26. MSFT is covered by Josh Baer. ORCL is covered by Sanjit Singh. GOOGL, META, and AMZN are covered by Brian Nowak.
Source: Company filings and Morgan Stanley Research.

Exhibit 4: ...while purchase commitments reach close to $1tn



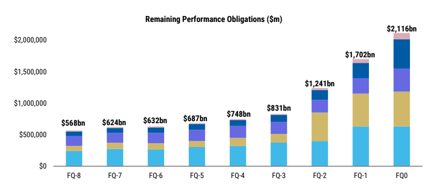
Note: Based on fiscal years. ORCL and MSFT 1Q26 = 3FQ26. MSFT is covered by Josh Baer. ORCL is covered by Sanjit Singh. GOOGL, META, and AMZN are covered by Brian Nowak.
Source: Company filings and Morgan Stanley Research.

Exhibit 5: AI ecosystem has become increasingly interconnected....



Note: MSFT, and CRWV are covered by Josh Baer. ORCL is covered by Sanjit Singh. NVDA and AMD are covered by Joseph Moore. AMZN is covered by Brian Nowak. DIS is covered by Sean Diffley. Data as of 3/19/26.
Source: Company Data, Morgan Stanley Research.

Exhibit 6: .... creating customer concentration within the >$2tn of RPO reported by major AI players



Note: GOOGL, META, and AMZN are covered by Brian Nowak. CRWV, and MSFT are covered by Josh Baer. ORCL is covered by Sanjit Singh.
Source: Company Filings, Morgan Stanley Research.

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AI Ecosystem

We are entering a record-setting capital cycle

Hyperscalers are pouring billions into the AI build-out, with finance lease usage pushing capital intensity even higher, well past the dot-com record. Capex estimates continue to revise higher, while sales revisions lag.

As investment grows, depreciation is set to rise

A sizable increase to depreciation expense could pressure margins if non-depreciation costs don't decline or if sales don't revise higher. But before these investments become expenses, much of the capital will sit in Construction in Progress (CIP), thus delaying earnings and margin impact.

Billions in lease and purchase commitments are fueling the AI buildout

Hyperscalers are securing capacity amid surging demand through leases and purchase commitments, creating a timing gap between monetization and supplier payment, leading to a rise in DPO among hyperscalers.

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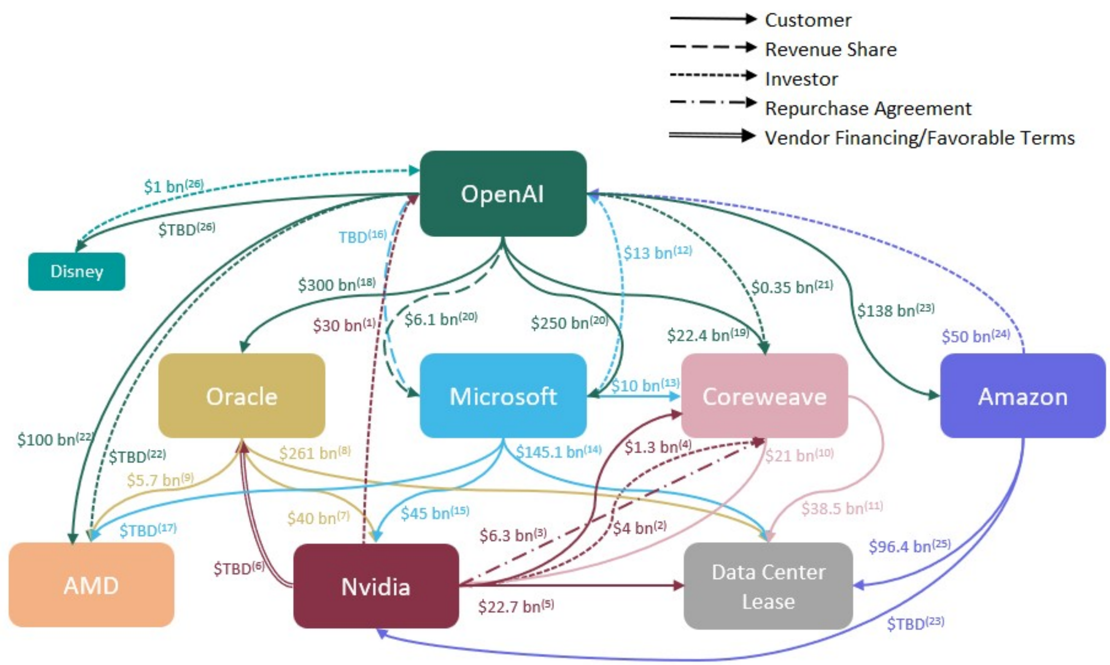
AI Ecosystem Capital Flows



Read more: [AI: Mapping the AI Ecosystem (22 Mar 2026)]()

Note: ORCL, MSFT, and CRWV are covered by Keith Weiss. NVDA and AMD are covered by Joseph Moore. AMZN is covered by Brian Nowak. DIS is covered by Sean Diffley. Data as of 3/19/26.

Source: Company Data, Morgan Stanley Research.

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Capital Intensity

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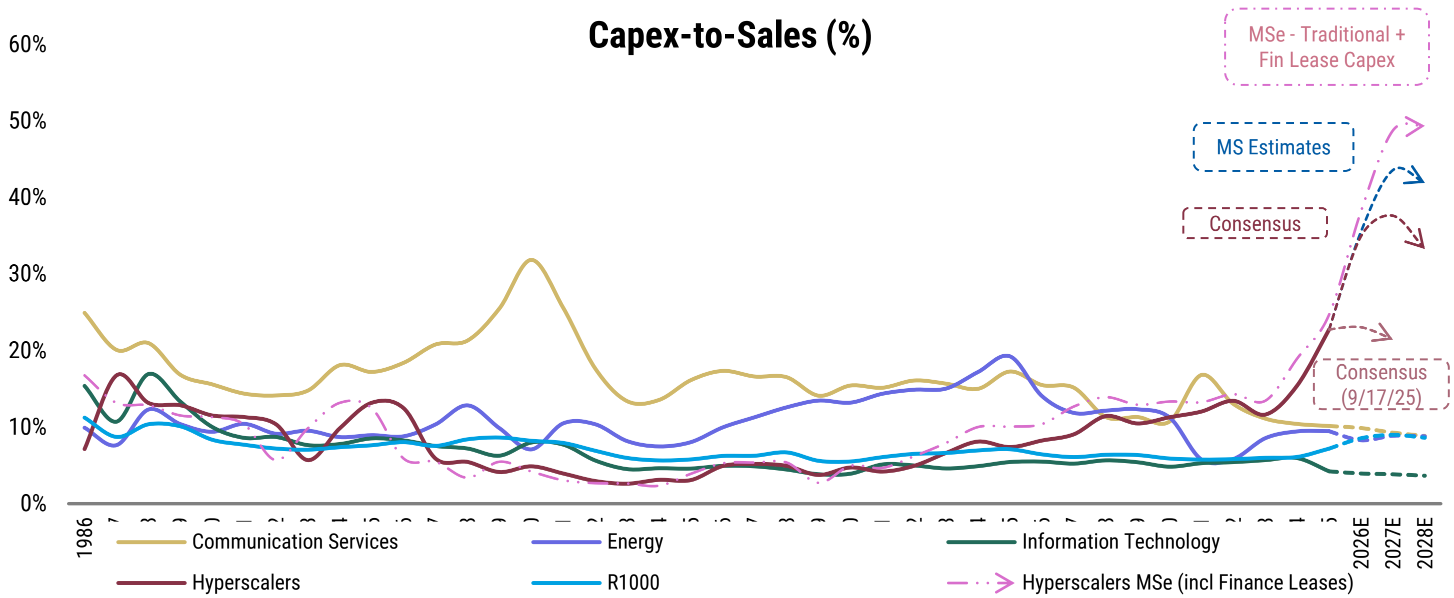
AI Boom Poised to Overtake Dot-Com Peak Levels

- MSe forecasts the AI capex cycle will **surpass peak capital intensity level seen during the dot-com era** (32%), reaching 36%, 44%, and 42% in 2026-2028.



Read more: [AI: Capital Intensity Enters a New Era (26 Feb 2026)]

Note: The hyperscalers group includes AMZN, GOOGL, META, MSFT, & ORCL and are excluded from the other sector groups. Only MSFT and ORCL finance lease estimates are included. R1000 excludes Real Estate and Financials.

Source: Morgan Stanley Research and FactSet

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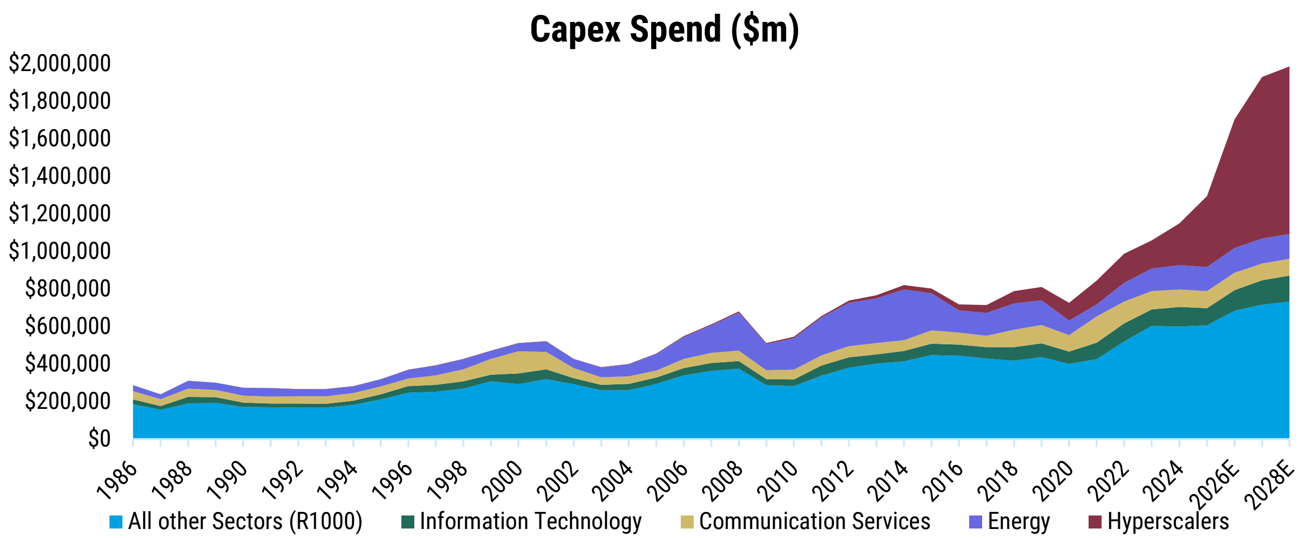
Hyperscalers Capex Spending Trajectory

- Hyperscalers accounted for **>150% of US Large-Cap capex growth in 2025** and are expected to represent **~40% of index capex in 2026**, doubling from 2024 levels.
- Other AI enablers (Energy, Industrials, etc.) will likely drive total AI capex north of 50% of aggregate capex.



Read more: [AI: Capital Intensity Enters a New Era (26 Feb 2026)]

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Source: Morgan Stanley Research and FactSet.

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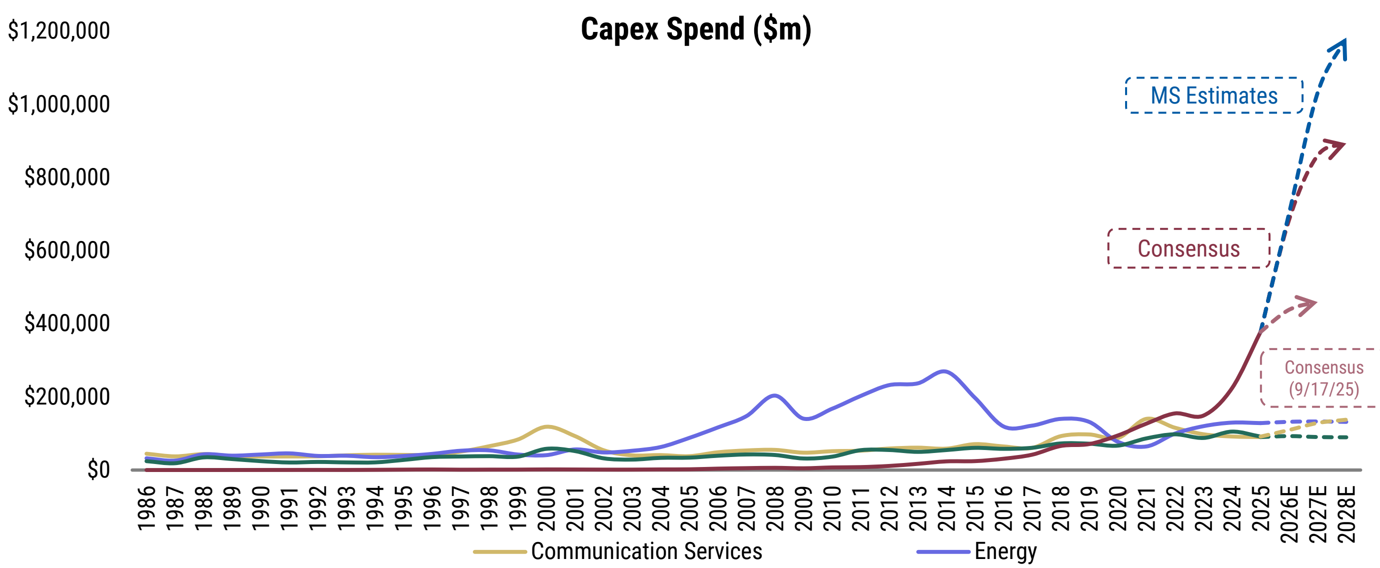
Capex Revisions Climb Higher…

- Consensus raised capex estimates by an **aggregate of ~$900bn in 2026-27 vs 9 months ago.**
- MSe hyperscaler spend forecasts align with consensus in 2026, but **diverge in 2027 and 2028 by 20% and 32%,** respectively.



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Note: The hyperscalers group includes AMZN, GOOGL, META, MSFT, & ORCL and are excluded from the other sector groups. R1000 excludes Real Estate and Financials. Estimates are based on consensus.

Source: Morgan Stanley Research and FactSet.

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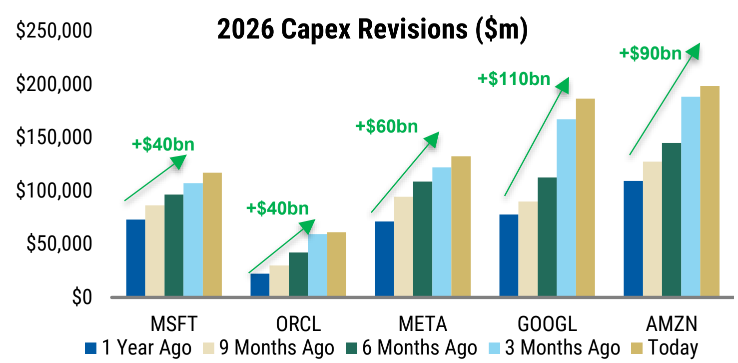
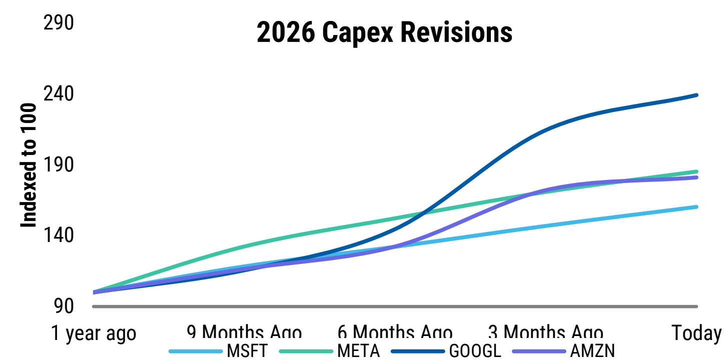
Capex Revisions Climb Higher…

- GOOGL consensus capex revisions for 2026 have increased 139% vs one year ago (June 2025), with META and AMZN up 85% and 81%, respectively.
- ORCL had largest jump with **revisions up 175% from one year ago.**





Read more: [AI: Capital Intensity Enters a New Era (26 Feb 2026)]

Note: GOOGL, META, and AMZN are covered by Brian Nowak, MSFT is covered by Keith Weiss. ORCL excluded to preserve scale. Latest data as of 6/4/26.
Source: Company filings and Morgan Stanley Research.

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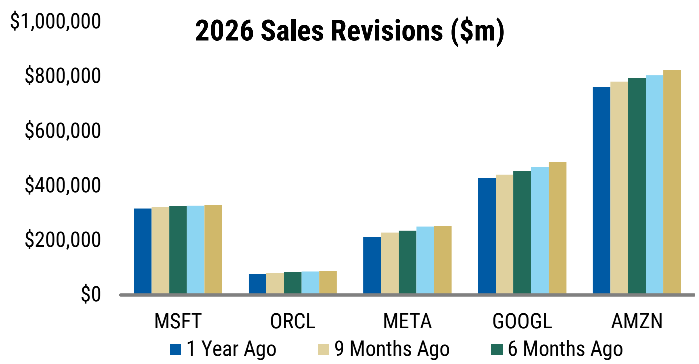
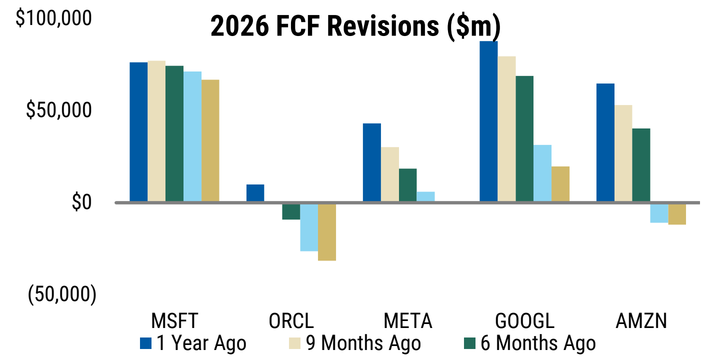
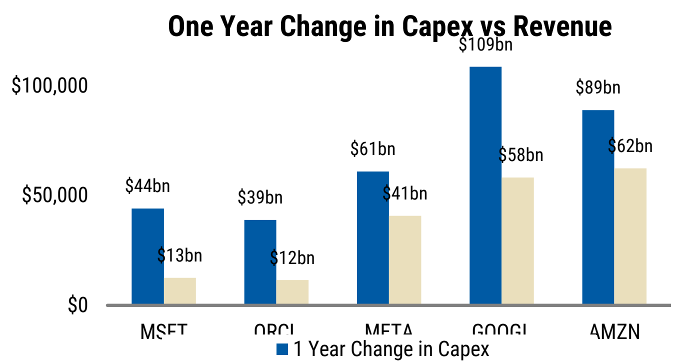
…Outpacing Sales and FCF Revisions







Read more: [AI: Capital Intensity Enters a New Era (26 Feb 2026)]

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Source: Morgan Stanley Research and Factset.

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Lease & Purchase Commitments

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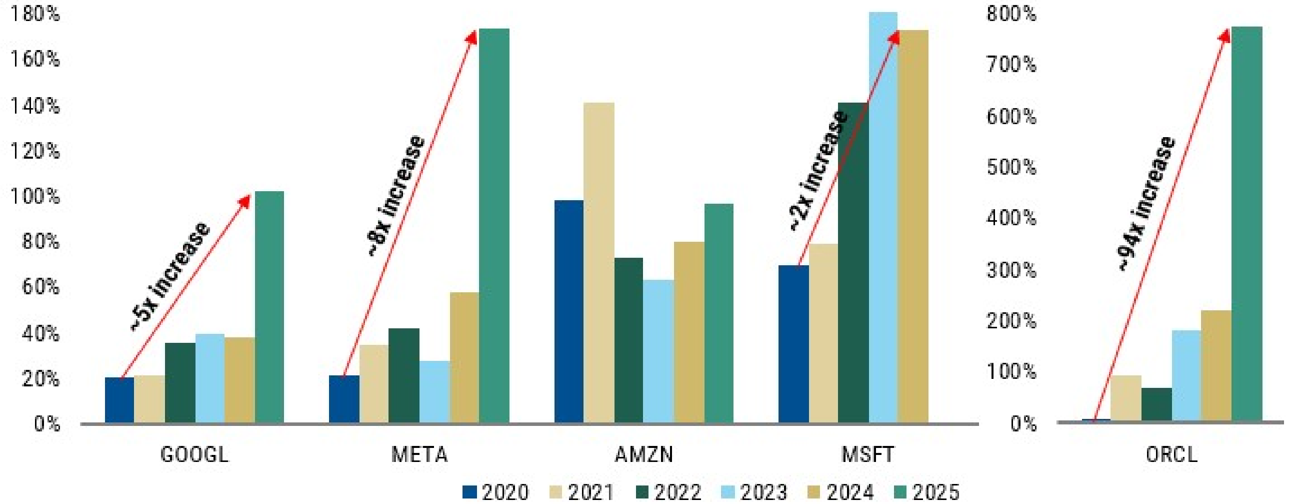
Lease & Purchase Commitments as % of NTM OCF



Read more: AI: Off Balance Sheet, On the Hook (15 Apr 2026)

Note: ORCL and MSFT figures are included in the prior annual period (2024 =FY 2025). MSFT 2025 = 2FQ26. ORCL 2025 = 3FQ26. GOOGL, META, and AMZN are covered by Brian Nowak, MSFT and ORCL is covered by Keith Weiss.
Source: Morgan Stanley Research, Company Filings, and Visible Alpha estimates.

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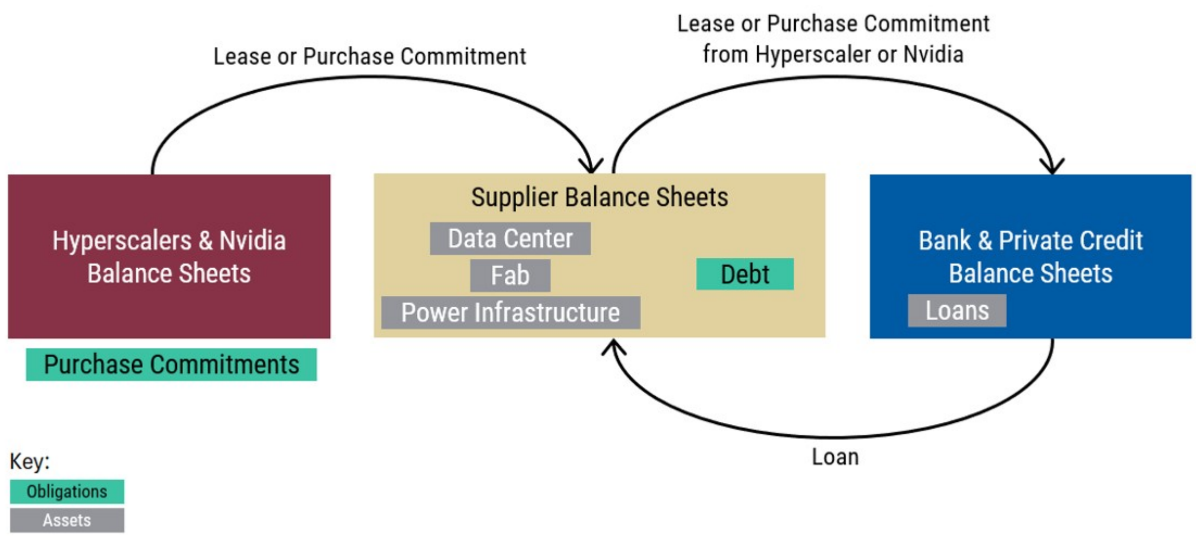
Suppliers Using Commitments from Hyperscalers & Nvidia to Access Financing



Read more: [AI: Off Balance Sheet, On the Hook (15 Apr 2026)]

Source: Morgan Stanley Research.

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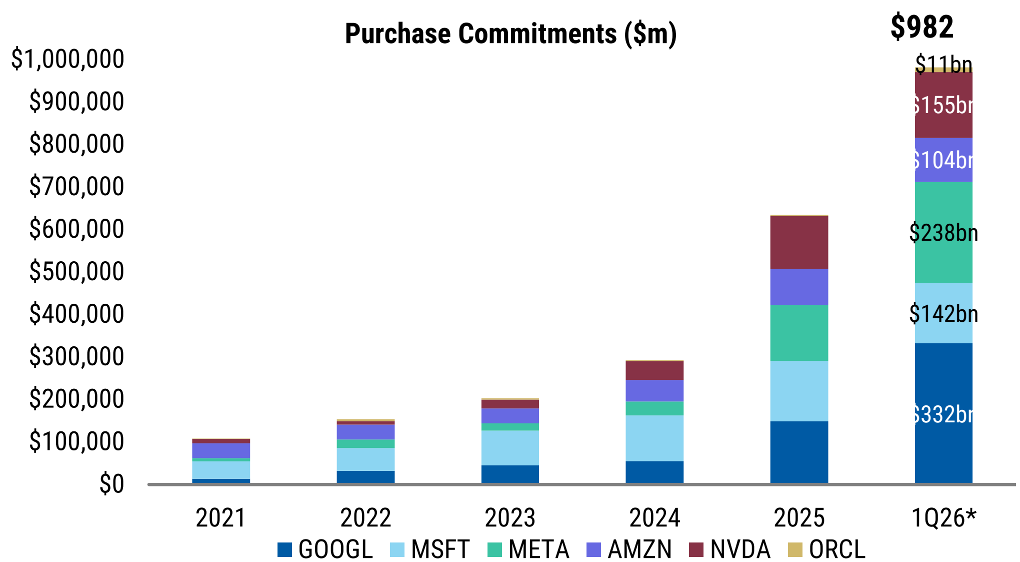
Hyperscaler/Nvidia Purchase Commitments close to ~$1T

- Unless a company expects a loss on these contracts, these obligations **remain off-balance sheet until the underlying goods or services are delivered** and a payable is recognized.



Read more: **AI: Off Balance Sheet, On the Hook (15 Apr 2026)** **and** **AI: Capital Intensity Enters a New Era (26 Feb 2026)**

Note: Based on fiscal years. ORCL 1Q26 = 3FQ26. MSFT 1Q26 = 2025 given no quarterly disclosure. GOOGL, META, and AMZN are covered by Brian Nowak, MSFT and ORCL are covered by Keith Weiss. NVDA is covered by Joe Moore.
Source: Morgan Stanley Research and Company Filings

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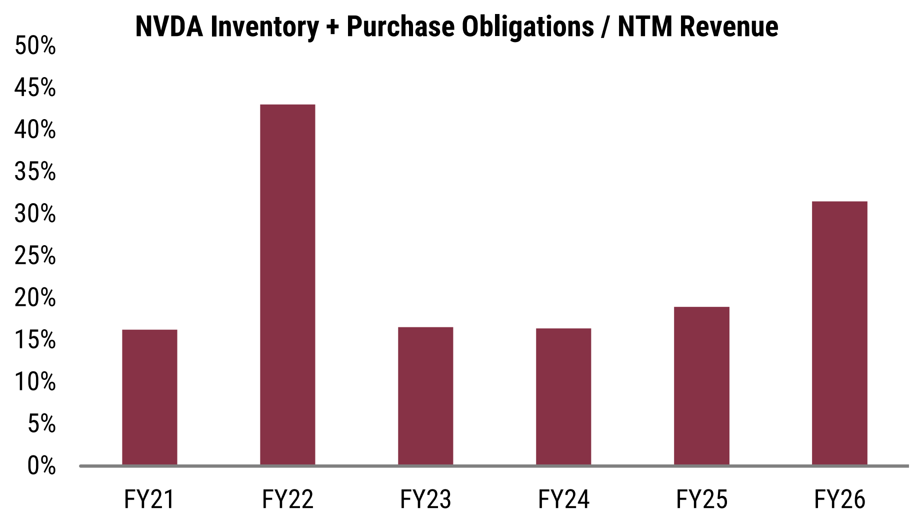
Nvidia Inventory Purchase Commitments

- Nvidia's inventory and inventory purchase commitments **have increased from 15–20% historically, to ~32%** of consensus FY27 revenue.



Read more: [AI: Off Balance Sheet, On the Hook (15 Apr 2026)]()
Note: NVDA is covered by Joe Moore.
Source: Company Filings, Factset, and Morgan Stanley Research. NTM estimate as of 4/10/25.

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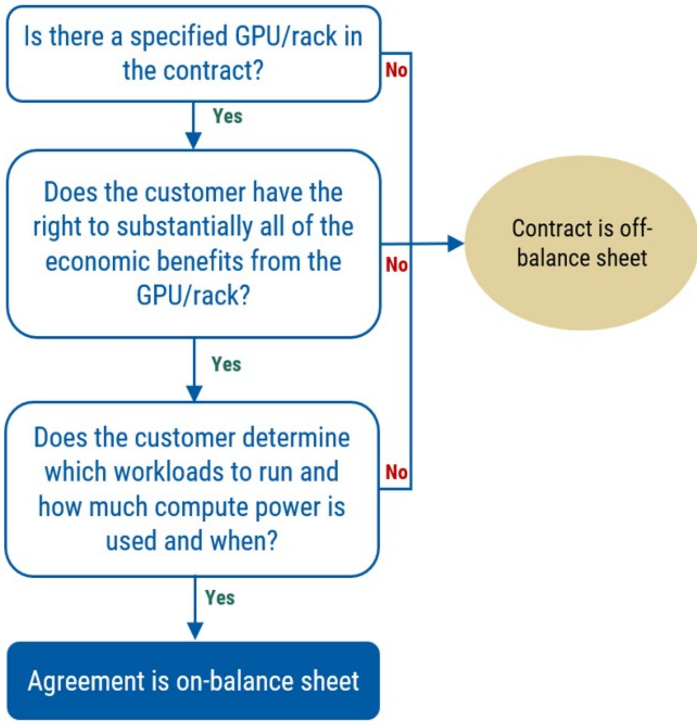
Compute Capacity Arrangements – *Does the Deal Contain a Lease?*



- As of their latest filings:
  - **Nvidia**: disclosed **$30bn** of cloud service agreement commitments
  - **Oracle**: disclosed **$10bn** of cloud capacity arrangements
  - **Meta**: did not quantify but said that its **$238bn** of contractual commitments include third-party cloud capacity arrangements

Read more: AI: Off Balance Sheet, On the Hook (15 Apr 2026)

Source: Morgan Stanley Research.

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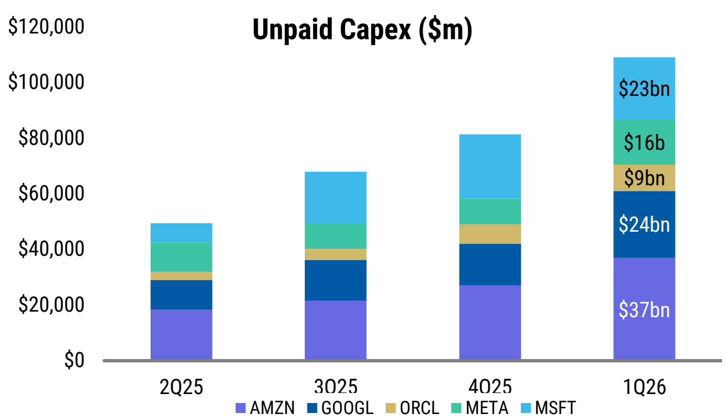
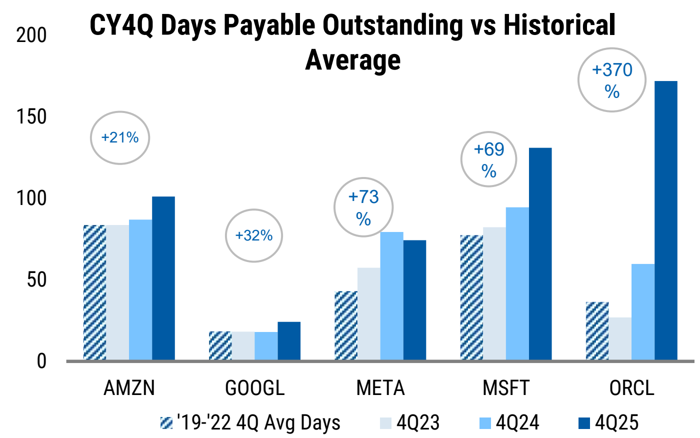
Purchases Driving up Accounts Payable among AI Players

- Unpaid capex embedded in accounts payable and accrued expenses now **totals ~$110bn across major AI players.**
- Timing gap from AI monetization to supplier payment is leading to growing DPO vs historical averages.





Read more: AI: Hyperscaling Payables (10 Feb 2026)
Note: GOOGL, META, and AMZN are covered by Brian Nowak. MSFT and ORCL are covered by Keith Weiss. ORCL's quarters based on its FYE (1Q26 = 3FQ26).
Source: Company Filings, Morgan Stanley Research.

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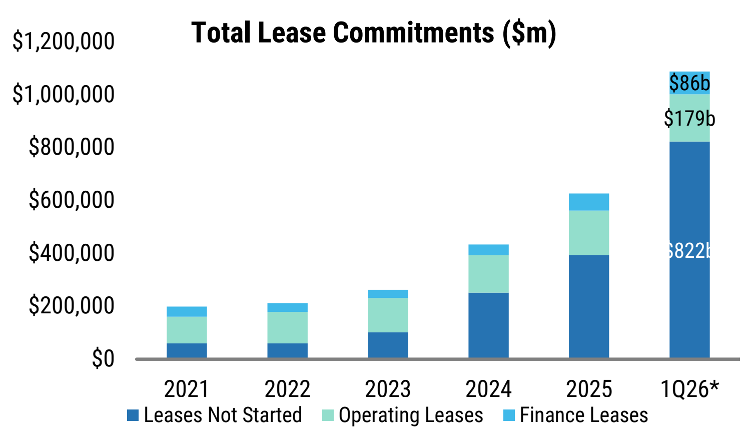
On/Off Balance Sheet Leases across Hyperscalers

- Common types of lease payments that **are not** included in the lease liability:
  - Leases Signed but Not Yet Started
  - Variable Lease Payments
  - Renewal Options
  - Residual Value Guarantees
  - Third-Party Lease Backstops & Guarantees



Read more: AI: Off Balance Sheet, On the Hook (15 Apr 2026) **and** AI: Leasing The Future (11 Nov 2025)

Note: Leases not started are undiscounted. Based on fiscal years. ORCL and MSFT 1Q26 = 3FQ26. GOOGL, META, and AMZN are covered by Brian Nowak, MSFT and ORCL are covered by Keith Weiss.
Source: Morgan Stanley Research and Company Filings.

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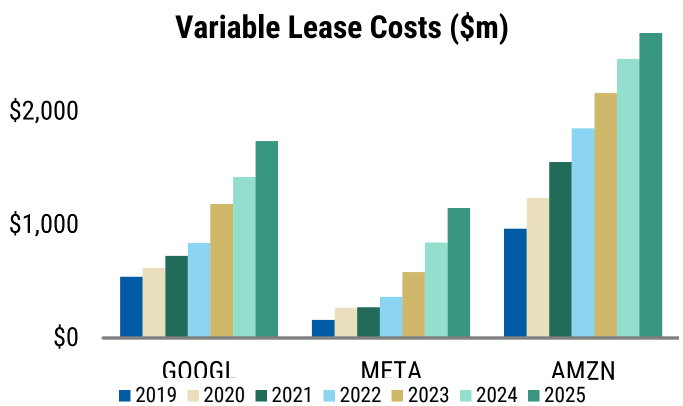
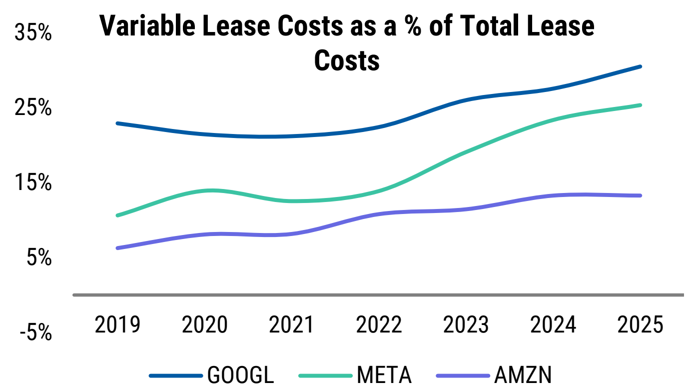
Off-Balance Sheet Example: *Variable Lease Payments*

- Variable lease payments exceed **10% of lease costs at AMZN, 25% at META, and 30% at GOOGL** (MSFT and ORCL report none, likely due to materiality judgments).





Read more: **AI: Off Balance Sheet, On the Hook (15 Apr 2026)**
Note: GOOGL, META, and AMZN are covered by Brian Nowak.
Source: Company Filings and Morgan Stanley Research.

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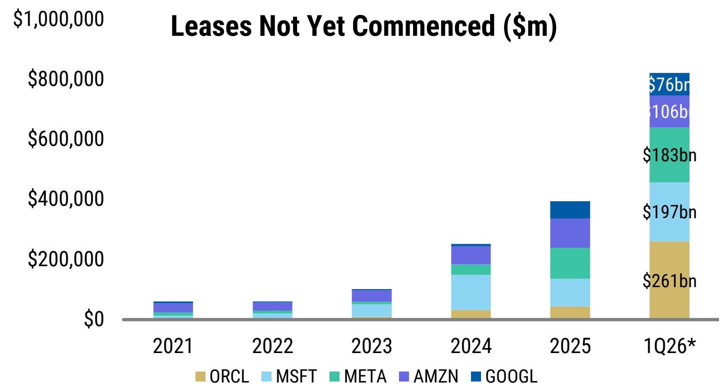
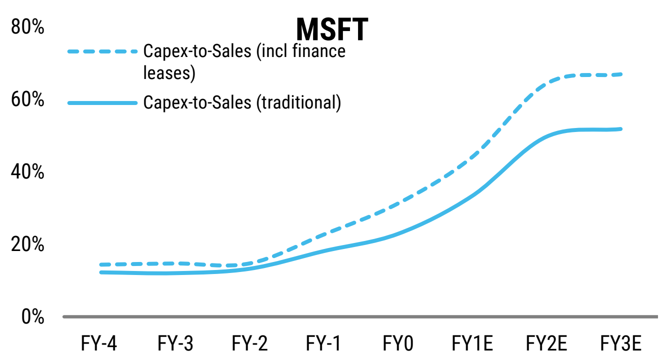
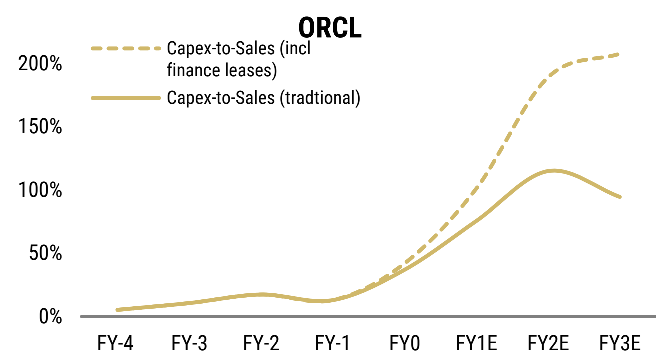
Finance Leases Pushing Up Capital Intensity

- **$800bn+ of lease commitments** highlight growing strategy by Hyperscalers to quickly expand AI build-out.
- MSFT's capex-to-sales incl finance leases could jump from 33% and 50% in FY26 and FY27 to **44% and 64%,** while ORCL's could jump from 76% and 115% to **101% and 189%,** over same period (MSe).







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Note: Based on fiscal years. ORCL and MSFT 1Q26 = 3FQ26. MSFT and ORCL are covered by Keith Weiss, GOOGL, META, and AMZN are covered by Brian Nowak.
Source: Company filings and Morgan Stanley Research.

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Depreciation

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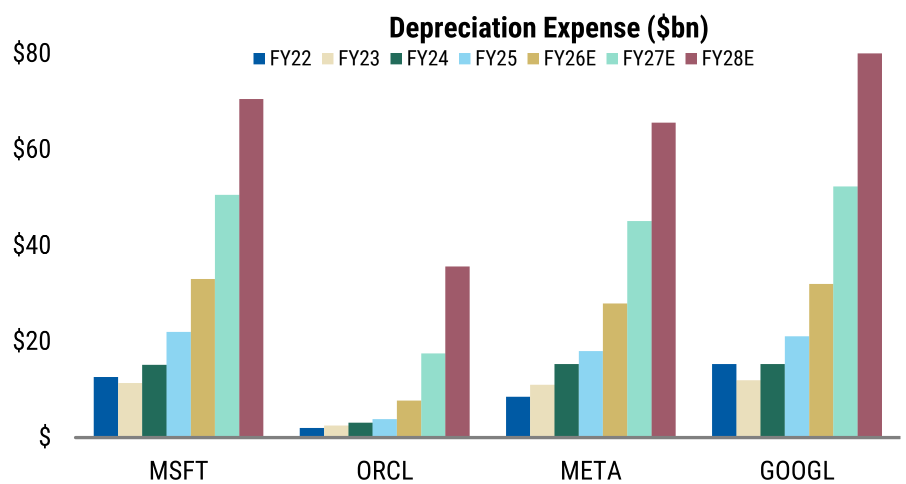
High Capital Investments → Significant Depreciation Expense

- We estimate cumulative depreciation for MSFT, ORCL, META, and GOOGL **could exceed $520bn over the next 3 years.**

Please reach out for our spreadsheet model that can be adapted to for your inputs.



Read more: **AI: Appreciating Depreciation Flows (28 Jan 2026)** **and** **AI: Depreciation Flows Update (26 Feb 2026)**

Note: Depreciation figures in 2026-2028 are estimates derived from our depreciation model, based on MS capex estimates. MSFT and ORCL are covered by Keith Weiss. META and GOOGL are covered by Brian Nowak.

Source: FactSet, Company filings, and Morgan Stanley Research.

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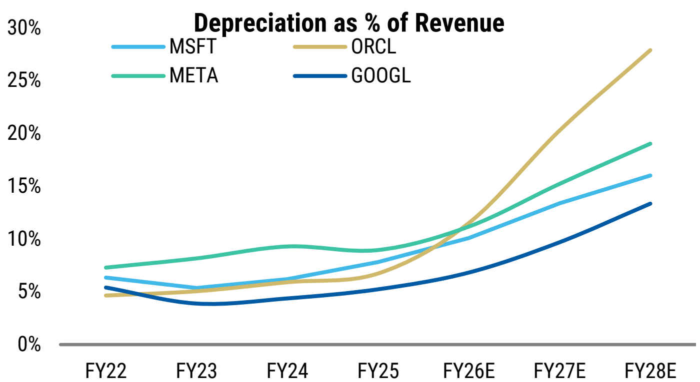
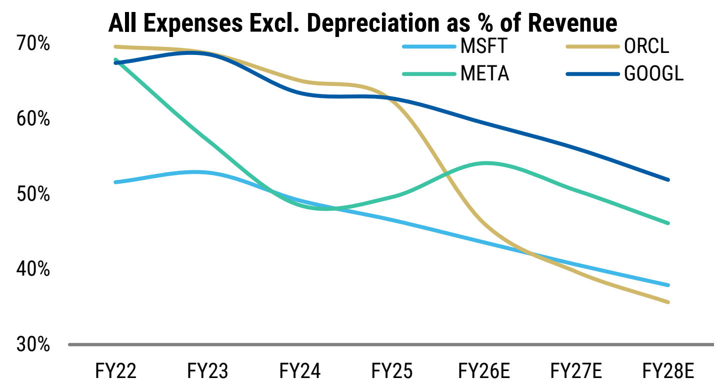
Depreciation Impact on Margins

- As depreciation becomes a larger % of revenue, all other expenses need to decline to achieve margin expectations.
- ORCL's depreciation as a % of revenue could rise from **7% to 28% in FY28**, while META's could jump from **9% to 19%**, over the same period.





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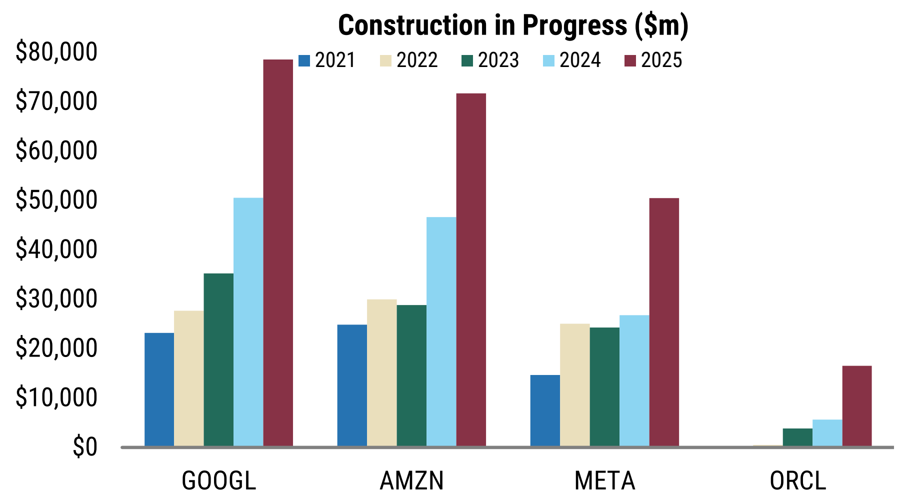
Construction in Progress Delaying Earnings Impact

- Rising CIP balances indicate the delay between the capital outlay and the expense hitting net income.
- **ORCL**, **META**, & **GOOGL's** CIP balances grew by ~200%, ~90%, & ~55% respectively, over the past year.



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Source: FactSet, Company filings, and Morgan Stanley Research.

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Remaining Performance Obligations (RPO)

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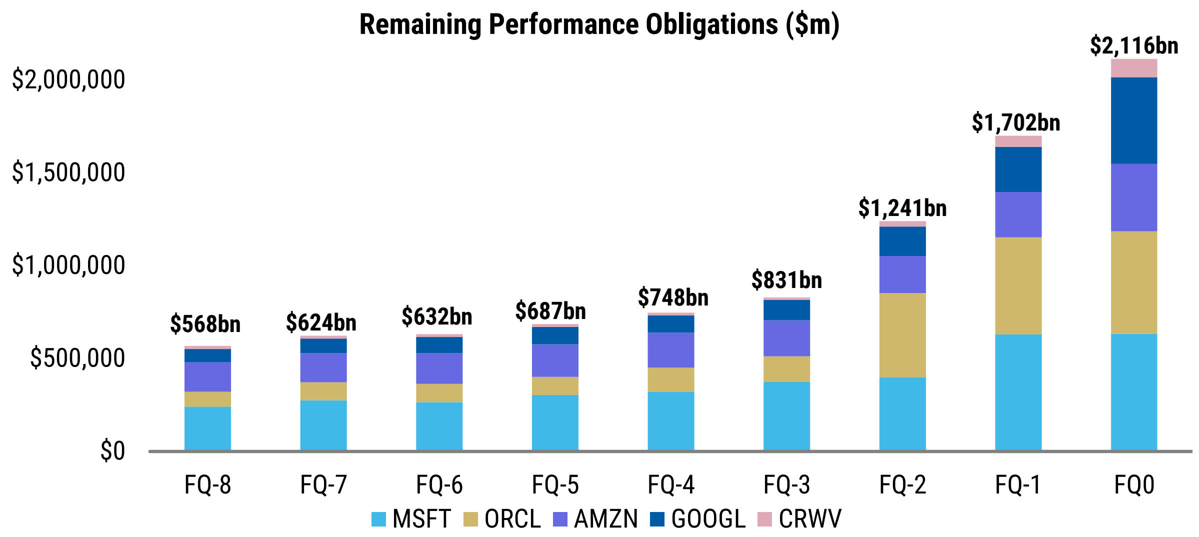
RPO Balances Rising among AI Players

- Large AI contracts have pushed up RPO balances with **ORCL's RPO balance up >300% yoy**, while **MSFT and GOOGL's balances have risen 97% and 406%,** over the same period.



Read more: [AI: RPO Revealed (21 Oct 2025)]

Note: GOOGL, META, and AMZN are covered by Brian Nowak. CRWV, MSFT and ORCL are covered by Keith Weiss.
Source: Company Filings, Morgan Stanley Research.

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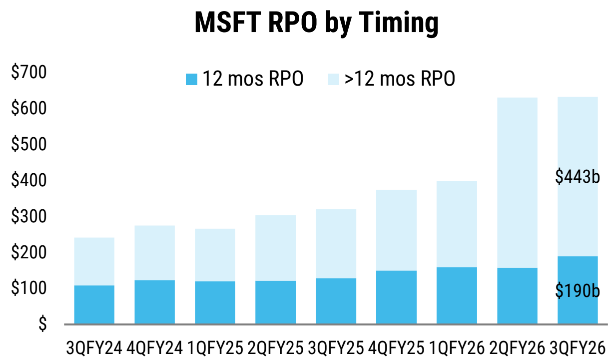
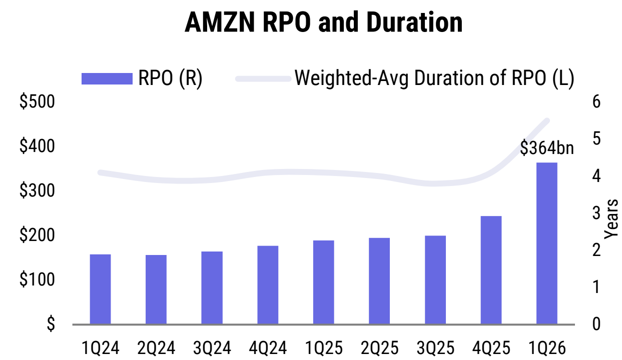
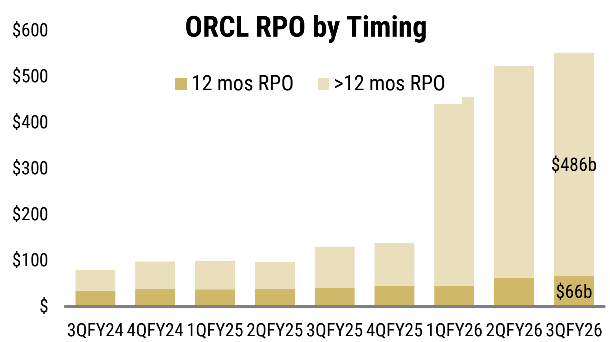
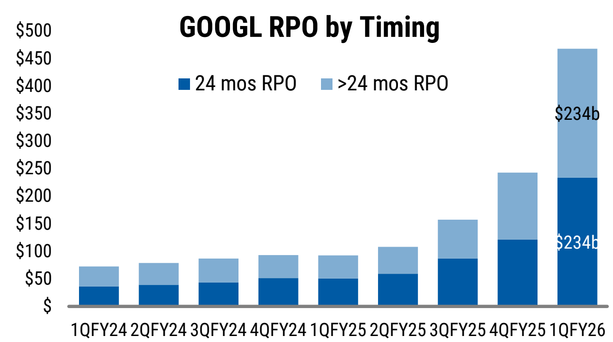
RPO Duration Differences









Read more: [AI: RPO Revealed (21 Oct 2025)]

Note: GOOGL, META, and AMZN are covered by Brian Nowak. CRWV, MSFT and ORCL are covered by Keith Weiss.
Source: Company Filings, Morgan Stanley Research.

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(as of May 31, 2026)

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	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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